Flight Evidence: Averages and Fair Claims

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Tracking #: 2026-11-11-NAT-SCI-ROCKET-EVIDENCE-44

Fictional controlled launch data (feet; three trials per design)

Rocket	Trial 1	Trial 2	Trial 3
X: short fins	7	9	8
Y: long fins	8	11	8
Conditions	Same launch line	Same straw	Same room

1. Calculate X mean and range with the exact numbers and feet labels.

2. Calculate Y mean and range; show each operation, not only the answers.

3. Compare the means and ranges. Name which model averaged farther and which varied more.

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4. Explain why the claim ’long fins guarantee the best launch’ is unsupported by these three trials.

5. Plan a repeat study with five trials per model: state the variable, the measure, TWO controls and what evidence would change your conclusion.